Executive Summary

Nassau Candy Factory Reallocation & Shipping Optimization

Project Overview

I built this project to find a practical way of checking factory assignments before making a change. The system uses historical Nassau Candy order data to predict shipping lead time and compare different factory choices.

What the System Does

- Prepares the order data and creates machine-learning features.
- Tests three models for shipping lead-time prediction.
- Finds the best model based on the evaluation results.
- Groups route performance using clustering.
- Simulates all five available factories for a selected product.
- Calculates predicted lead-time improvement.
- Shows confidence and risk information.
- Provides the final results through a Streamlit dashboard.

Model Results

Model	MAE	RMSE	R^2
Linear Regression	181.422	182.490	0.5291
Random Forest	135.405	160.466	0.6359
Gradient Boosting	159.638	169.039	0.5960

Random Forest was selected as the final model because it had the lowest RMSE. It also gave the best MAE and R^2 of the three models tested.

How the Recommendation Works

For a selected product, region and ship mode, the system checks the current factory and compares it with the other four factories. If another factory has a lower predicted lead time, the system can recommend the alternative. If not, it keeps the current factory.

Dashboard

The Streamlit dashboard allows the user to select a product, region and ship mode. It also includes a speed-versus-profit priority slider and displays factory results, confidence, risk, profit margin and the final decision.

Important Limitation

The available dataset does not contain factory-specific manufacturing costs for a hypothetical reassignment. Because of this, I have not claimed a specific profit increase from changing factories. The current dashboard treats hypothetical profit impact as stable and points out this limitation.

Final Recommendation

- Use the tool to compare factory options before making a reassignment.
- Pay attention to both predicted improvement and confidence.
- Do not change a factory when the model does not show an improvement.
- Add real factory-level cost data before using the system for financial optimization.
- Continue collecting data so the model can be retrained and improved.

Conclusion

The project is a working prototype that combines machine learning with what-if analysis. It gives a more useful decision-support view than simply showing historical shipping numbers. The next major improvement would be adding real factory cost, capacity and transportation information so that both shipping performance and profitability can be optimized together.